AUTOMOBILE ENGINEERING · PART 7 OF 11

Hybrids, CNG and Ethanol

By Isagi · 9 November 2026 · All specs as of November 2026 (manufacturer brochures and listings)

TL;DR

Between a pure petrol car and a pure EV sit three alternatives. **Hybrids** add an electric motor and a small battery to a petrol engine, giving big efficiency gains in city traffic without any charging. **CNG** burns compressed natural gas: low fuel cost, with some loss of power and boot space. **Ethanol** blended into petrol (E20 today) is mostly a national energy policy that affects your car slightly. On claimed figures the Honda City hybrid and the Toyota Hyryder hybrid return about 27 to 28 km/l, about 57% to 61% more than the 17.4 km/l petrol Creta, although these are different cars.

1. The hybrid family

Type	What it adds	Can it drive on electric alone?	Examples
Mild hybrid	Small 48-volt motor assisting the engine	No	Mercedes C-Class (48V, up to 20 hp and 200 Nm boost)
Strong hybrid	Bigger motor and battery, can move the car alone	Yes, at low speed	Toyota Hyryder, Innova Hycross, Honda City e:HEV
Plug-in hybrid (PHEV)	Large battery you can charge from a socket	Yes, for tens of km	MG Hector PHEV*

Strong hybrids matter most to mass-market buyers because they need no plug. Their trick is to let the electric motor do the work the engine does worst: pulling away and crawling in traffic. The petrol engine then runs mostly at steady, efficient loads, and the motor recaptures braking energy that would otherwise be lost.

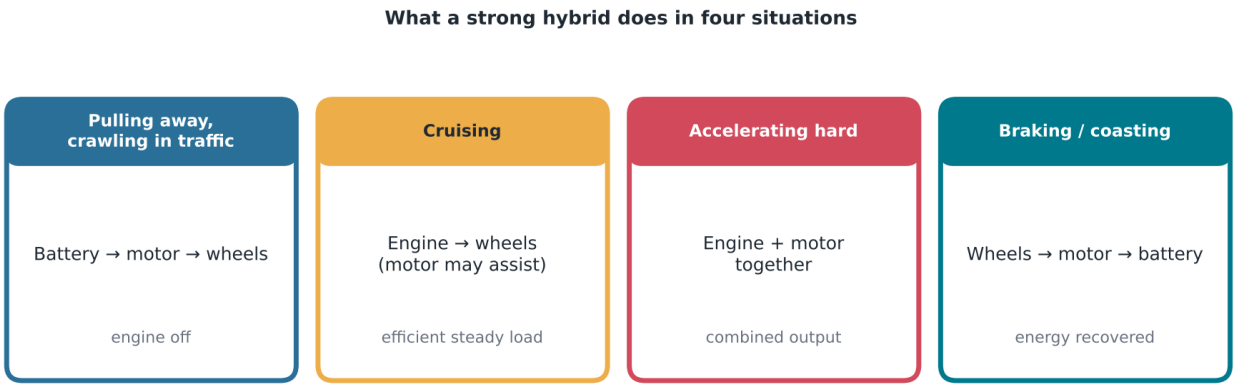


Fig 1. What a strong hybrid does in four common situations.

Both hybrids on sale in the 1.5-litre class use the Atkinson cycle, a variant in which the intake valve stays open longer so the engine uses less of the air it takes in each cycle, trading peak power for efficiency; the electric motor makes up the missing punch.* The numbers show how the two makers split the work:

	Toyota Hyryder hybrid	Honda City e:HEV
Engine	1.5 L, 3-cylinder	1.5 L, 4-cylinder (Atkinson cycle)
Engine power / torque	91 bhp / 122 Nm	97–98 bhp / 127 Nm

	Toyota Hyryder hybrid	Honda City e:HEV
Electric motor power / torque	79 bhp / 114 Nm	108–109 bhp / 253 Nm
Combined power	114–115 bhp	125–126 bhp
ARAI claimed efficiency	27.97 km/l	27.26 km/l

The City's motor is stronger than its engine and does most of the driving in many conditions; Toyota's system is slightly more efficient on paper. The Innova Hycross hybrid uses a 2.0-litre engine plus motor for about 184 bhp of system output, so the same idea scales up to a family MPV (Part 1).

2. CNG: cheap fuel, small penalties

CNG is natural gas stored under high pressure in a steel cylinder. The engine runs on it with a modified fuel system and usually can switch back to petrol. The Maruti Swift S-CNG shows both the benefit and the cost:

- **Efficiency:** 32.85 km/kg claimed (ARAI).
- **Power:** about 69–70 hp and 102 Nm, against 80.46 bhp and 111.7 Nm for the petrol engine, which is roughly 13–14% less power and 9% less torque.
- **Boot space:** the gas cylinder usually takes up luggage room; some makers fit twin smaller cylinders to keep more boot (for example Tata's iCNG range*).
- **Availability:** CNG stations exist mainly in some cities and along certain corridors, so check your area before buying.

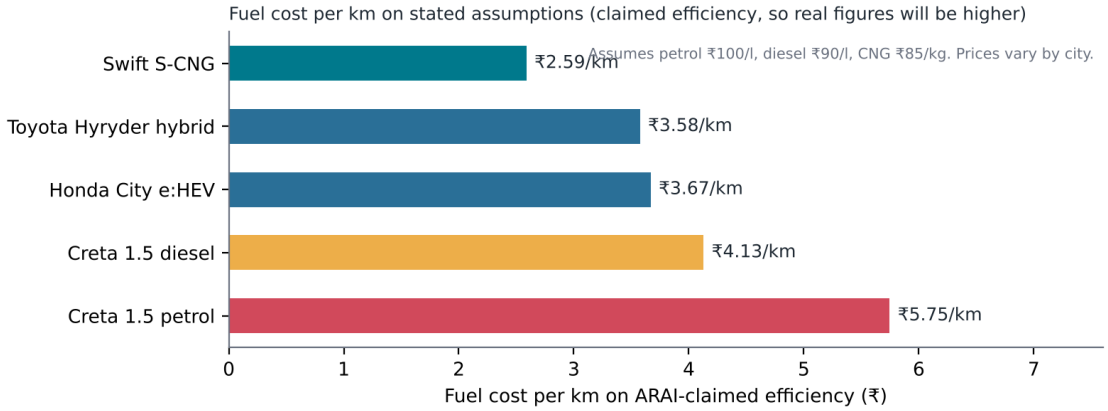


Fig 2. Fuel cost per km using claimed efficiency and the assumed prices shown. Different cars, so read the gaps as scale, not an exact comparison.

On these assumptions the CNG Swift costs under half as much per km as the petrol Creta. These are different cars, so compare like for like within one model before deciding; still, the gap is why CNG dominates taxi fleets and high-mileage family cars. The break-even logic of Part 6 applies again: divide the CNG price premium by the saving per km to see how many kilometres it takes to pay back.

3. Ethanol: E20 and flex fuel

Ethanol is alcohol made from sugarcane or grain. Petrol in India is now sold blended with 20% ethanol (**E20**).* Ethanol holds less energy per litre than petrol (roughly two-thirds), so E20 has about 7% less energy per litre than pure petrol, by this arithmetic: $0.8 \times 100 + 0.2 \times 66 \approx 93$.* A car not tuned for it can lose a few percent of km/l; newer engines are calibrated for E20 (Skoda, for example, announced E20 compliance for its 1.0 TSI engines).* Older cars may need compatible rubber hoses and seals over time.

Flex-fuel engines can run on any blend up to nearly pure ethanol. They are a policy bet to cut crude-oil imports and support farmers. Critics raise concerns about food-versus-fuel land use, water use and lower mileage; supporters point to import savings and lower tailpipe carbon.* Part 10 looks at how these policies shape the market.

4. Which one for whom

- **Mostly city driving, wants an automatic and low fuel bills with no charging:** strong hybrid.
- **High annual kilometres on a tight budget, with a CNG station nearby:** CNG.
- **Can charge at home and drives short distances:** EV (Part 6).
- **Long highway runs, heavy loads:** diesel or a plain efficient petrol still make sense (Part 2).

Summary

- Mild hybrids assist the engine; strong hybrids can drive on electric alone; plug-ins add a chargeable battery.
- A strong hybrid shines in city traffic because the motor handles low-speed work and recovers braking energy.
- The City e:HEV and Hyryder hybrid claim about 27–28 km/l, compared with 17.4 km/l for a petrol Creta.
- CNG cuts fuel cost per km sharply (the Swift S-CNG costs under half the petrol Creta per km on claimed figures, though they are different cars) but costs about 13–14% of power and some boot space on the Swift.
- E20 petrol carries about 7% less energy per litre; engines tuned for it lose little.
- Match the fuel to your kilometres, traffic and access to CNG or charging, not to the headline mileage.

*Unverified or general knowledge: PHEV example (MG Hector), Atkinson-cycle description, Tata iCNG twin-cylinder layout, E20 rollout status and compatibility remarks, ethanol energy content (about two-thirds of petrol, textbook value), the critics-and-supporters summary of flex-fuel policy and Skoda's E20 announcement are not checked against official documents. Specs: Carlelo (Hyryder and City hybrid ARAI 27.97 and 27.26 km/l, engine and motor figures), Autocar India (Swift CNG 32.85 km/kg, 70 hp, 102 Nm; Hycross hybrid), Wikipedia (Swift petrol figures). Fuel prices in Fig 2 are assumptions, not quotes.

See also: Part 6 (EVs) and Part 2 (mileage). Next: Part 8, Software, connectivity and driver assistance.